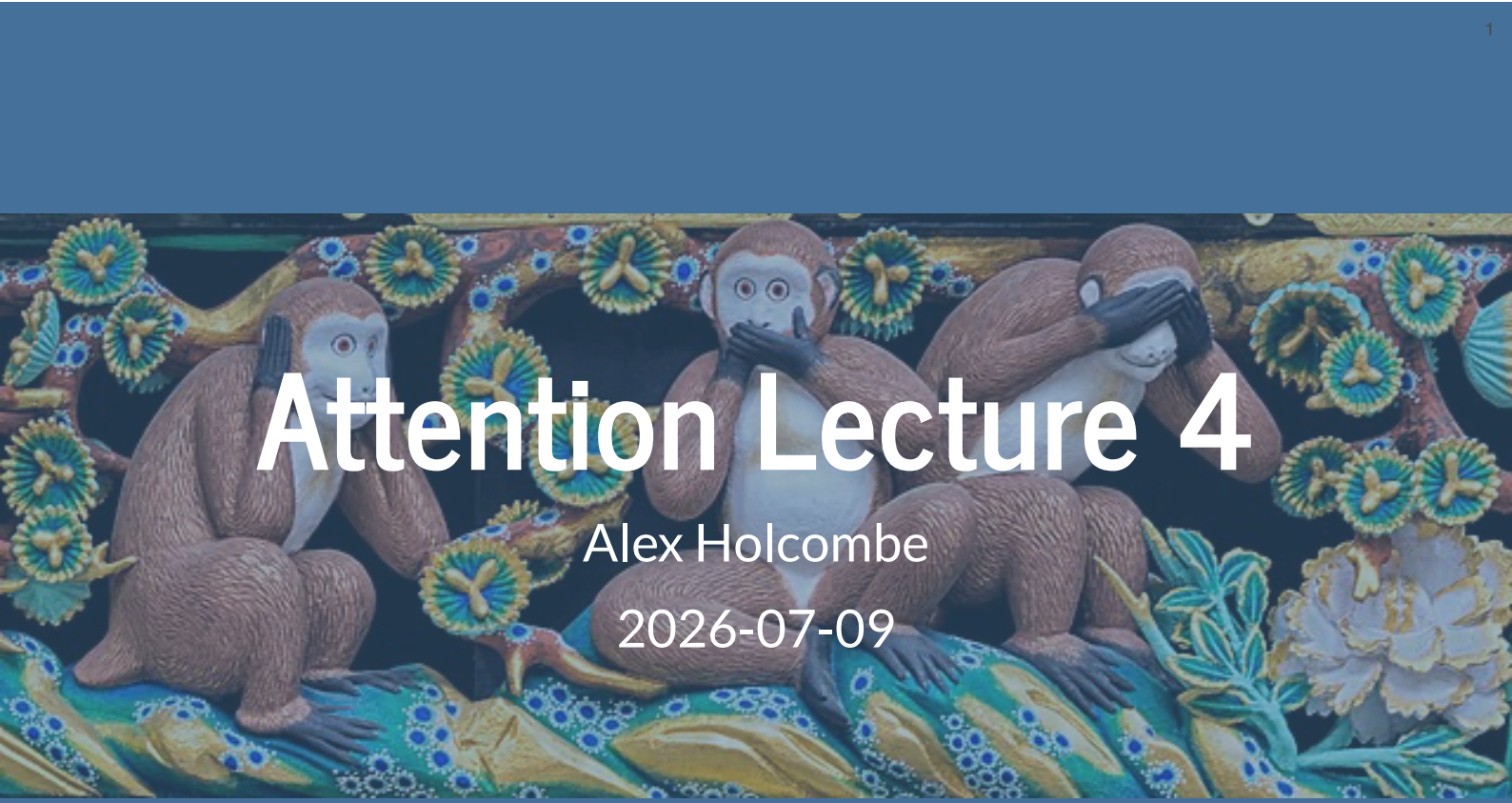


A colorful illustration of three brown monkeys in a jungle. The monkey on the left is looking at the viewer. The monkey in the middle is covering its mouth with its hands. The monkey on the right is covering its eyes with its hands. They are surrounded by green foliage and yellow flowers.

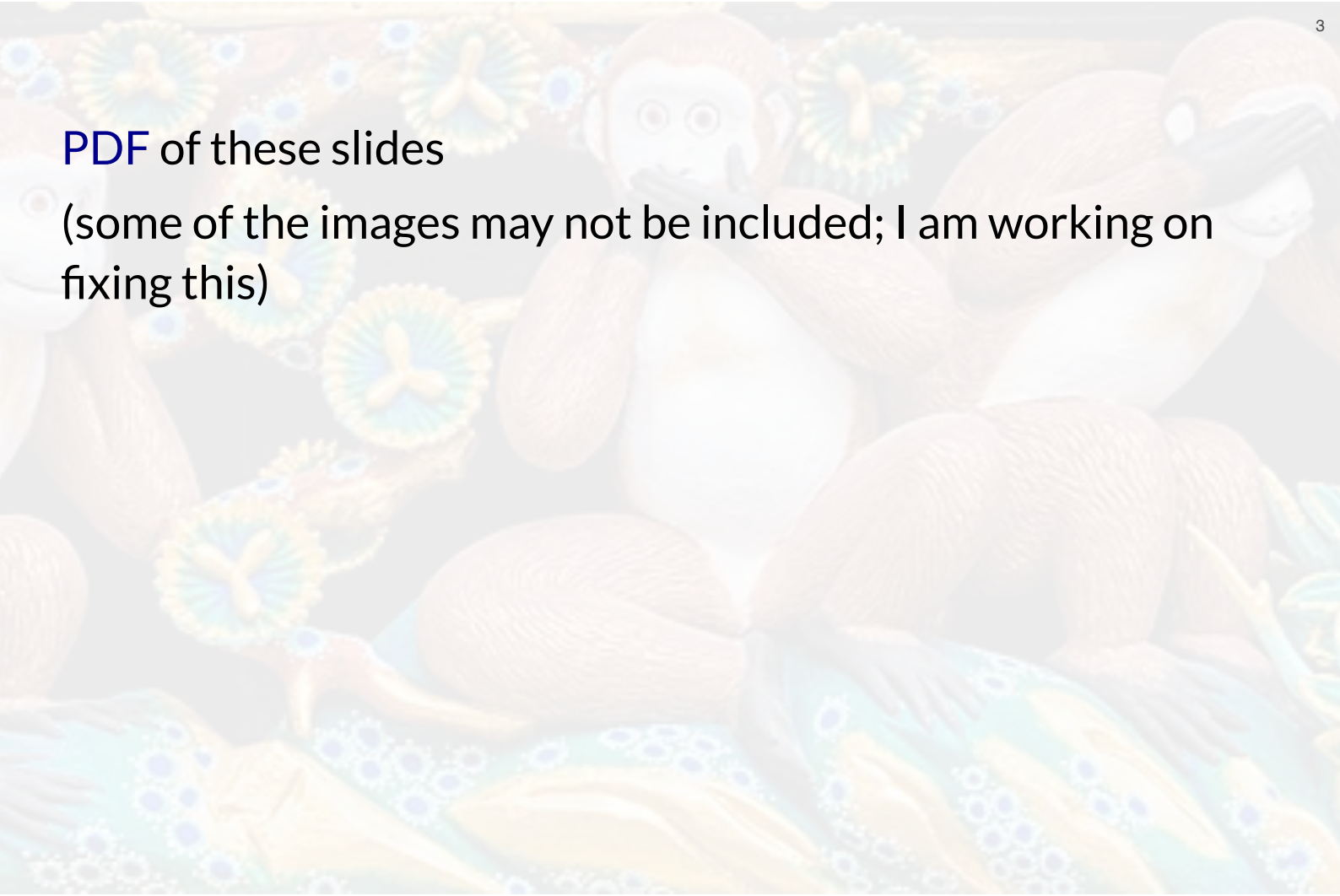
Attention Lecture 4

Alex Holcombe

2026-07-09

PDF of these slides

(some of the images may not be included; I am working on fixing this)

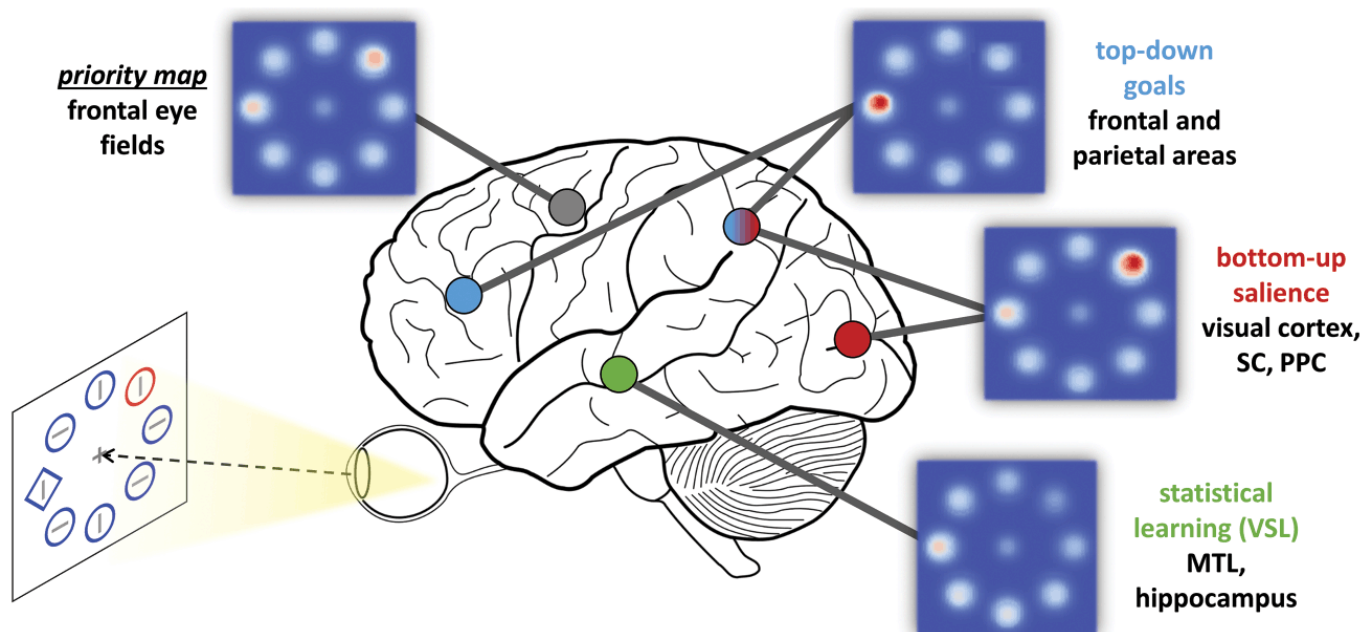


- **Distinguish the ways a stimulus can be selected**
 - Contrast *bottom-up* (*saliency-driven*) and *top-down* (*goal-directed*) attentional control.
 - Identify the types of top-down selection: spatial/location and feature selection.
- **Explain why feature selection is global (Ch. 7)**
 - State that attending to a feature (e.g., red) enhances that feature across the *entire* visual field, not just a restricted region.
 - Explain why you cannot simultaneously attend to red on the left and blue on the right.
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 - Describe how voluntary goals can supplement or override bottom-up saliency signals.
 - Give examples of when saliency alone is insufficient and top-down attention must step in.
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 - Define *parallel processing* as simultaneous processing across many locations in the visual field.
 - Explain how a unique feature value (e.g., an oddball color or orientation) generates a salient signal that pops out, driving bottom-up attention.
 - Distinguish *feature dimensions* (e.g., color, orientation, motion) from *feature values* (e.g., red, blue; leftward, downward).
- **Explain why identifying feature combinations is slow (Ch. 7, 8, Tutorial)**
 - Contrast flat (parallel) search slopes for single-feature targets with steep (serial) slopes for conjunction targets.
 - Describe why combining features requires focused attention and is capacity-limited.
- **Summarise Treisman's Feature Integration Theory (Ch. 9)**
 - Outline the FIT model: a pre-attentive stage that computes feature maps in parallel, followed by an attentive stage that binds features at a location.
 - Explain how *illusory conjunctions* arise when features are incorrectly bound without focused attention.
 - Apply FIT to predict whether a given visual search will be easy (parallel) or difficult (serial).

Lecture 4

- Location selection
- Feature selection
- Absence of combined selection (Ch.7), need for focused attention to combine features (Ch.9)

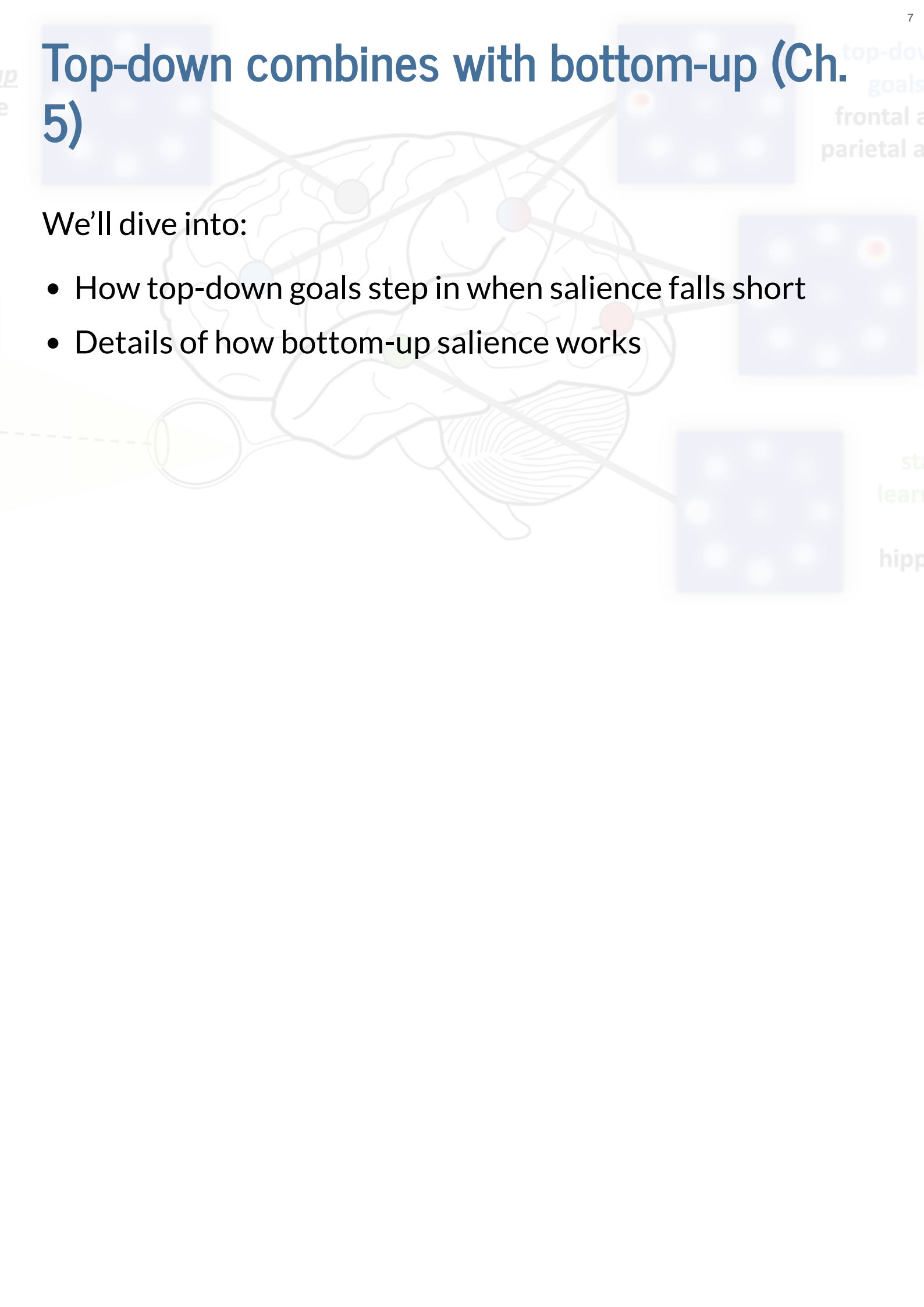
Top-down combines with bottom-up (Ch. 5)



Top-down combines with bottom-up (Ch. 5)

We'll dive into:

- How top-down goals step in when salience falls short
- Details of how bottom-up salience works



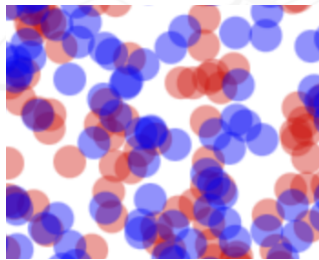
Ways to select a stimulus

- Bottom-up
- Top-down

- Spatial/location selection

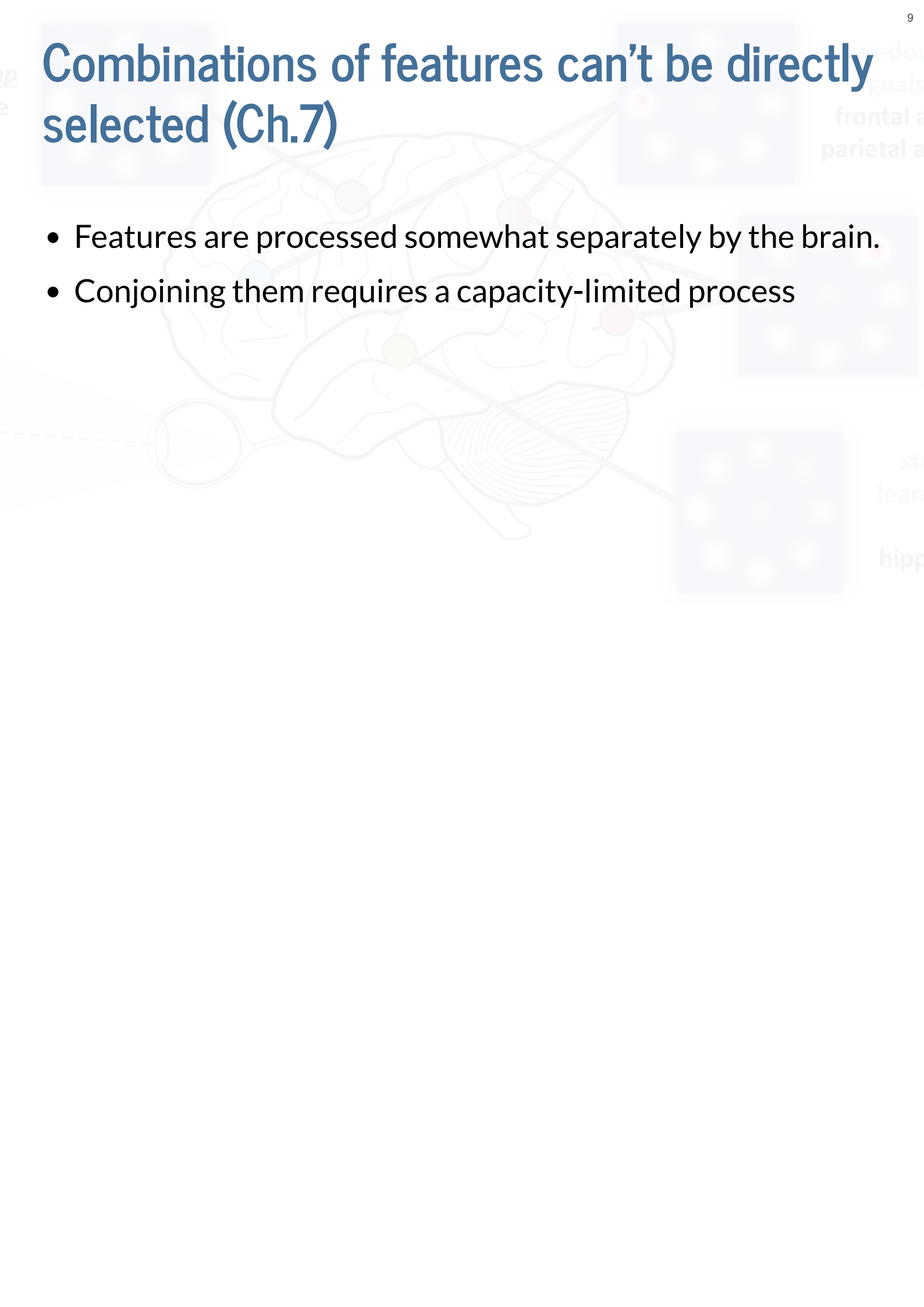


- Feature selection



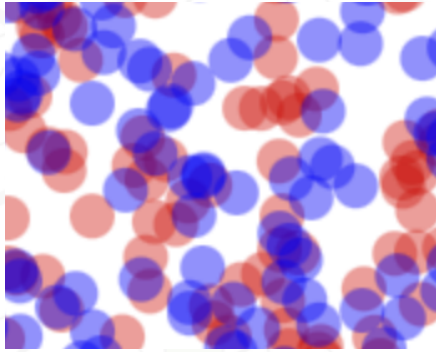
Combinations of features can't be directly selected (Ch.7)

- Features are processed somewhat separately by the brain.
- Conjoining them requires a capacity-limited process

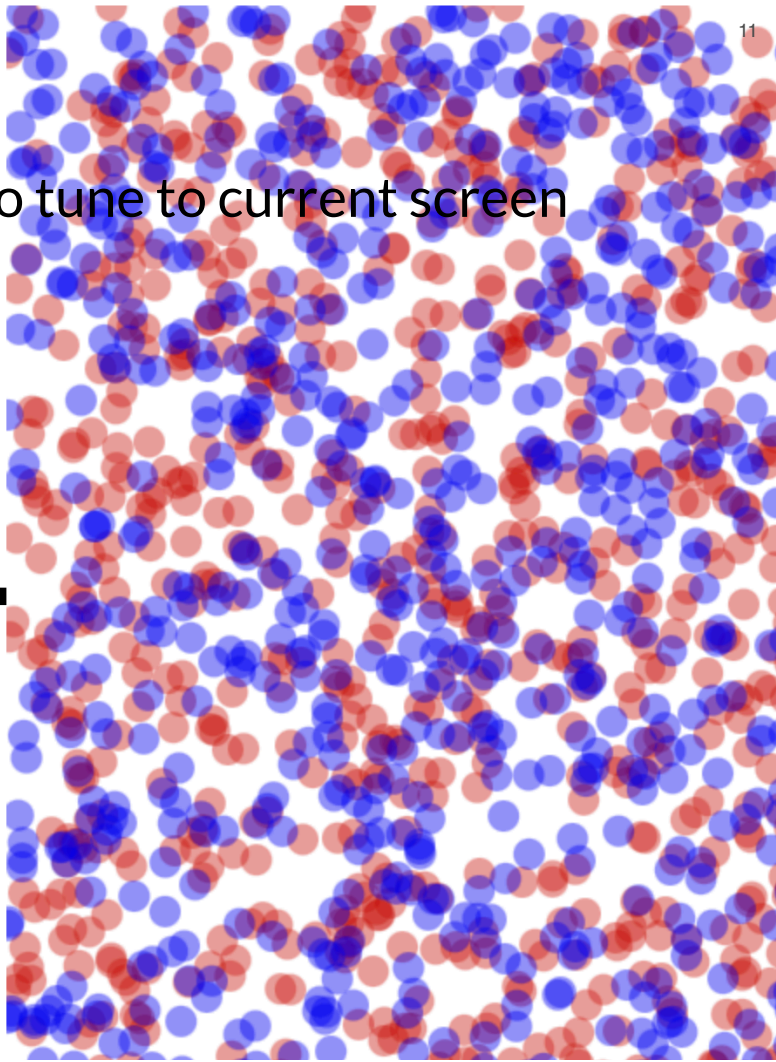
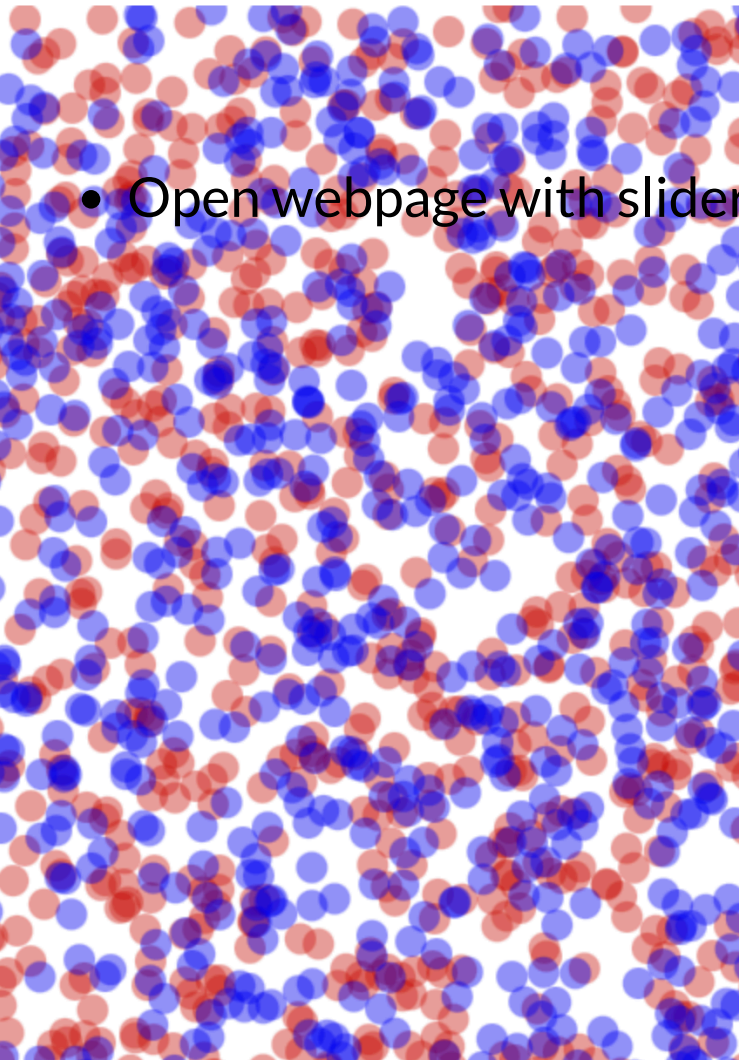


Feature selection is global (Ch.7)

Feature selection

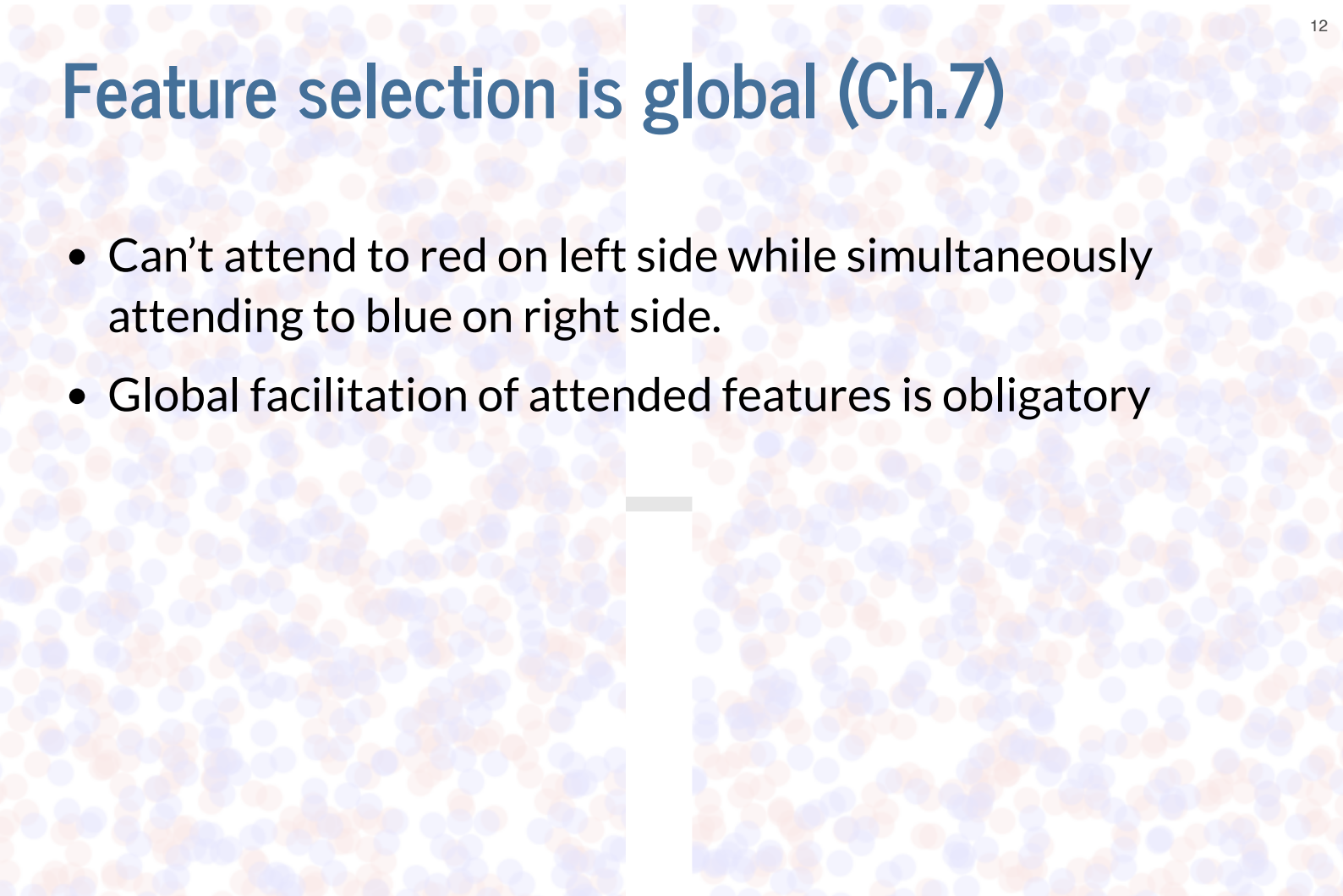


- Open webpage with sliders to tune to current screen

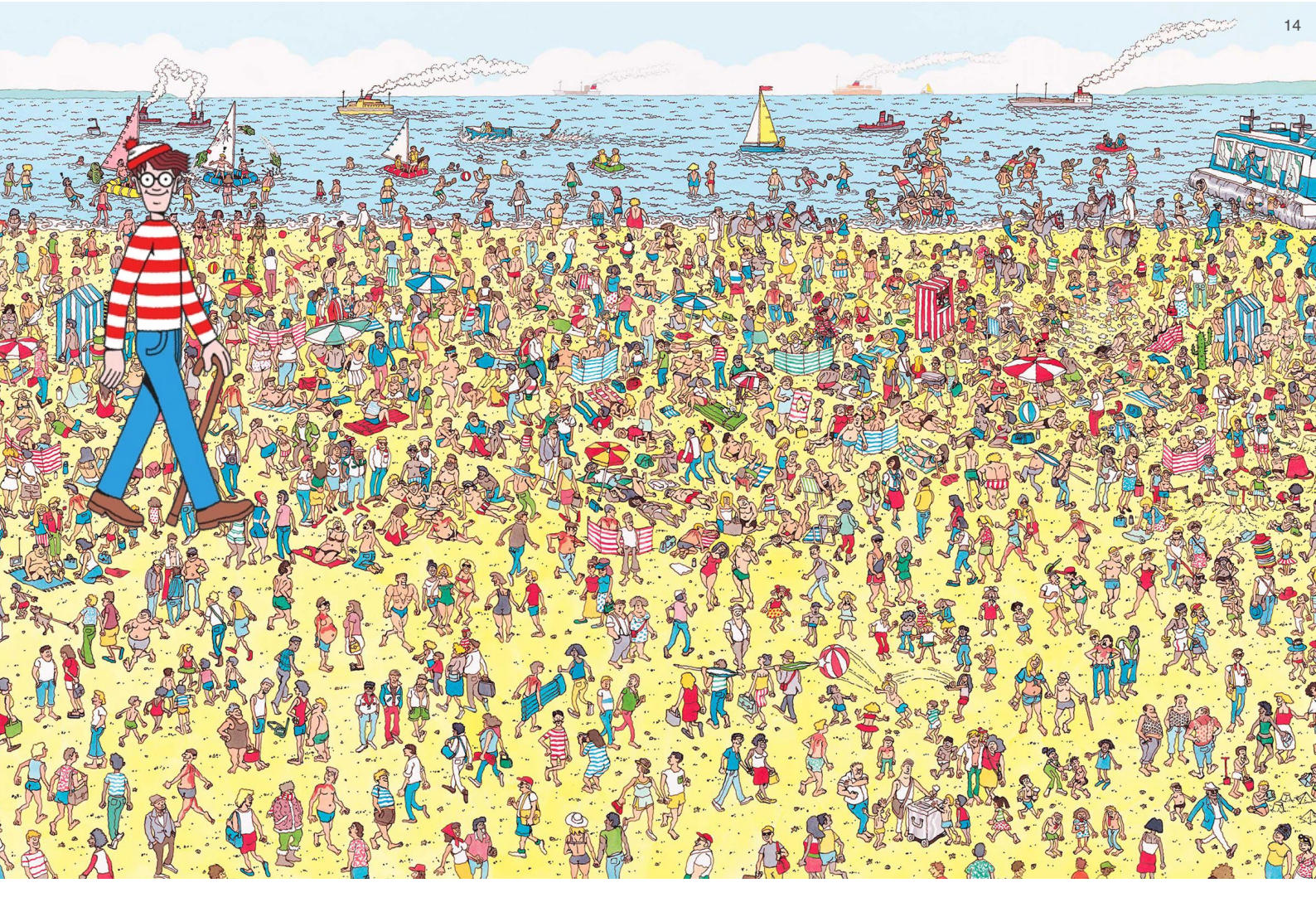


Feature selection is global (Ch.7)

- Can't attend to red on left side while simultaneously attending to blue on right side.
- Global facilitation of attended features is obligatory





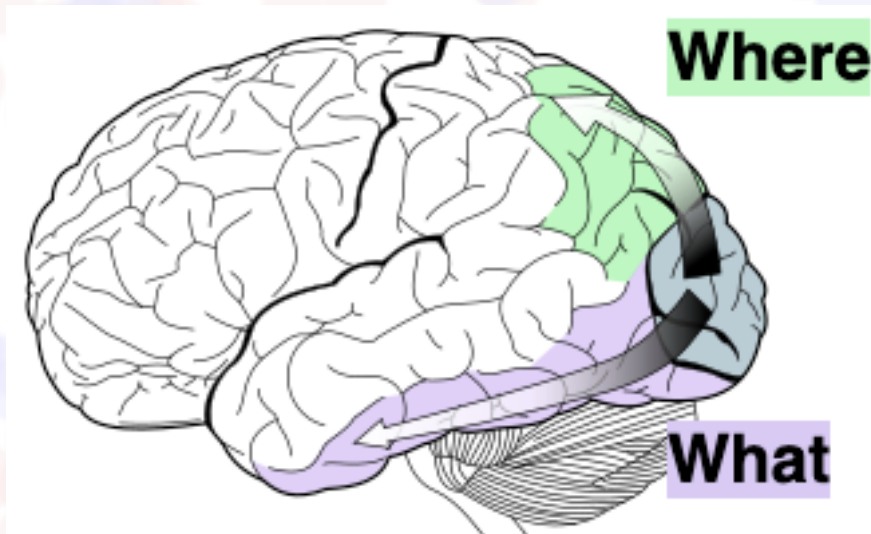


Feature selection is global

The *combination* of location and feature selection can't be done.

You can either select a simple feature everywhere (e.g., red objects), or select the stimuli in a particular region, but you can't combine those two abilities.

Global: Combining location and feature selection can't be done



Separation of what and where pathways.

- dorsal = parietal = where = location
- ventral = temporal = what = object recognition

Why are Where and What separated?

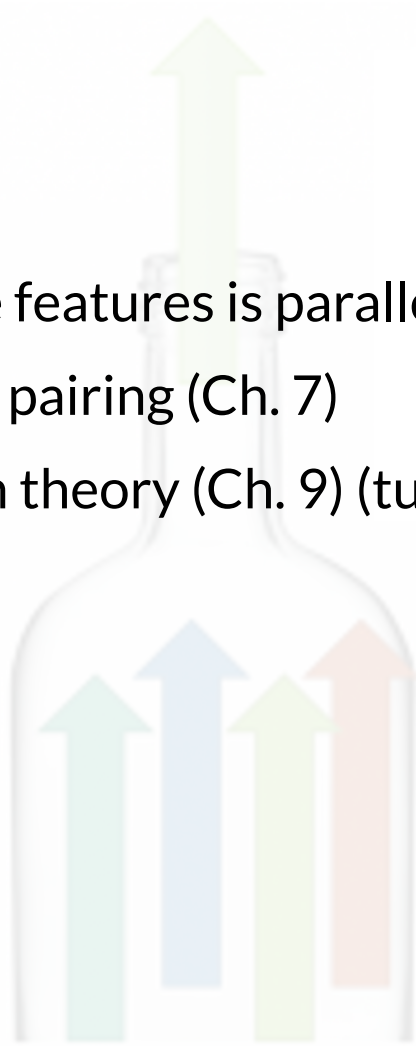
- Object recognition (What) doesn't need location information much
- Where is good for action, doesn't need object identity much

Han, Z., & Sereno, A. (2023). Is it always computationally advantageous to use segregated pathways to process different visual stimulus attributes separately? *Journal of Vision*. <https://doi.org/10.1167/jov.23.9.5020>

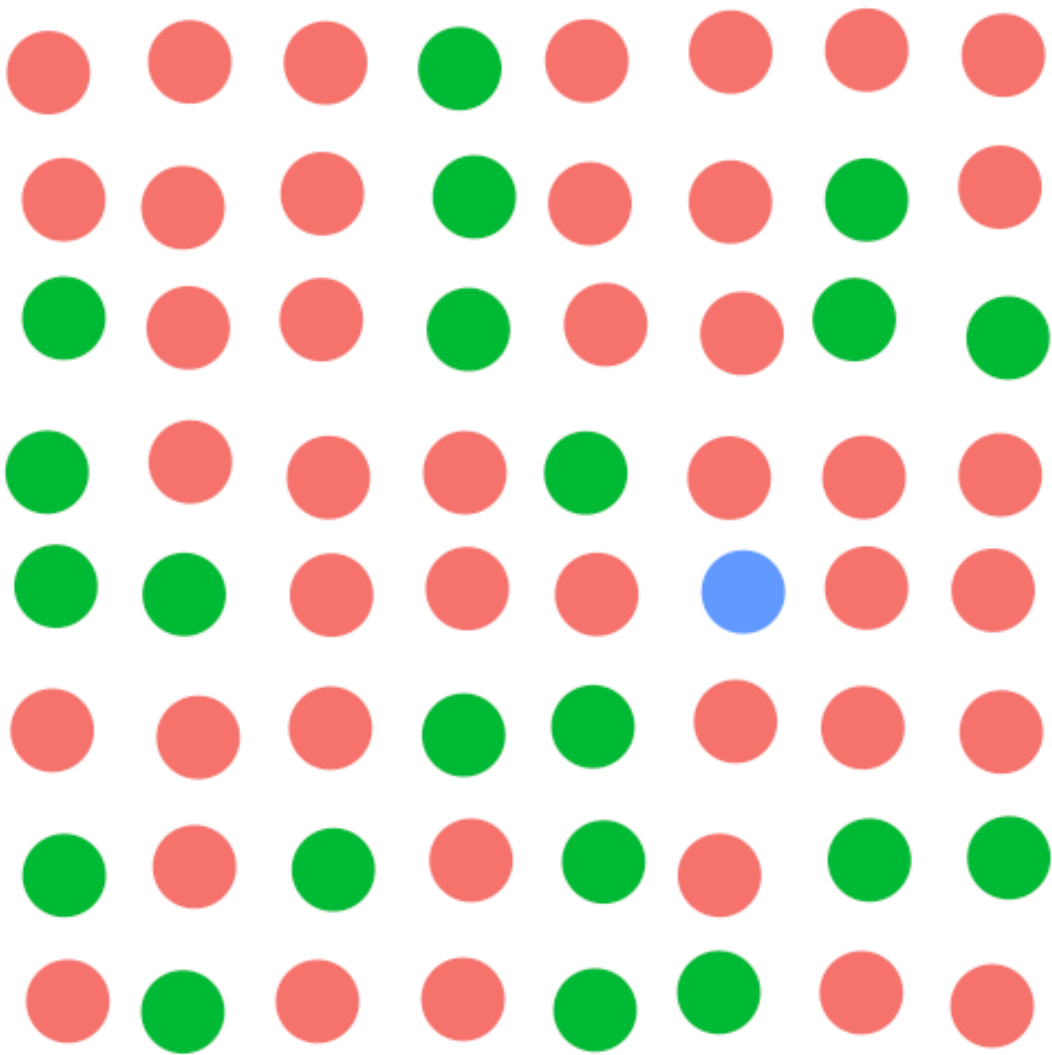


Lecture 4a

- Processing of some features is parallel (Ch. 5, Ch. 7)
- Absence of feature pairing (Ch. 7)
- Feature integration theory (Ch. 9) (tutorial)



Parallel feature processing (Ch.5,7)

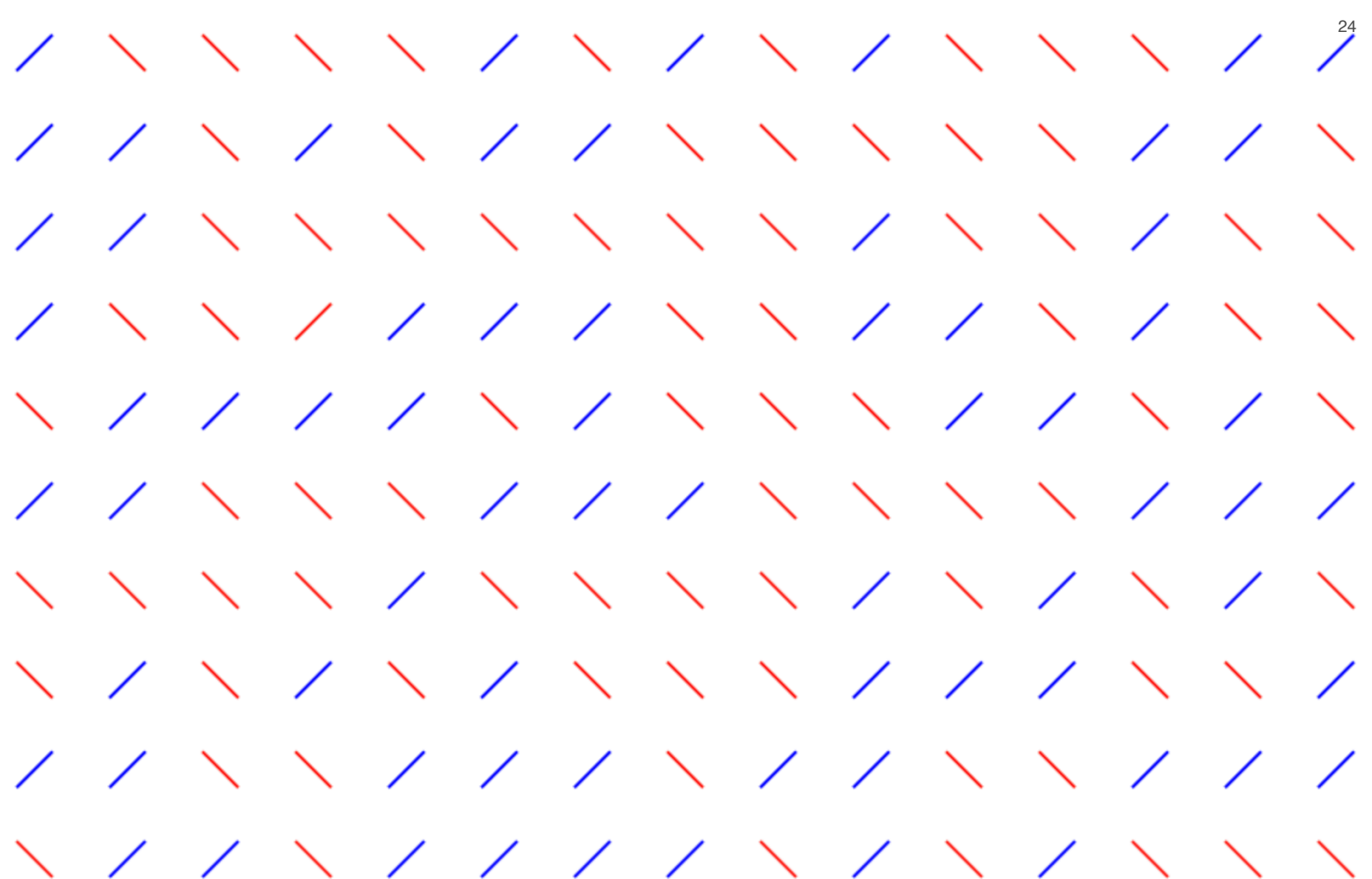


Parallel feature processing

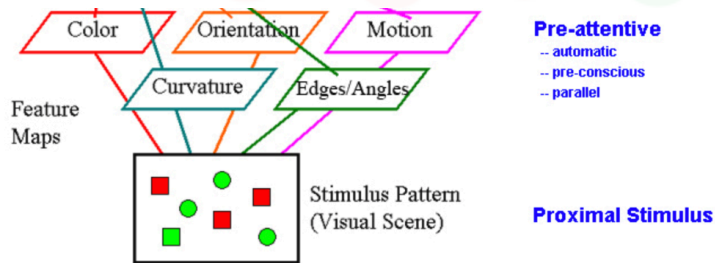
- Visual cortex detects color difference
 - Processes whole visual field simultaneously
 - Oddball color signal propagates (saliency)
 - Drives bottom-up attention
- Two reasons the previous slide is easy
 - If I tell you to look for blue, you activate blue things in your visual field
 - Difference in color from the others



Next, you need to find the line that's red and tilted to the right.

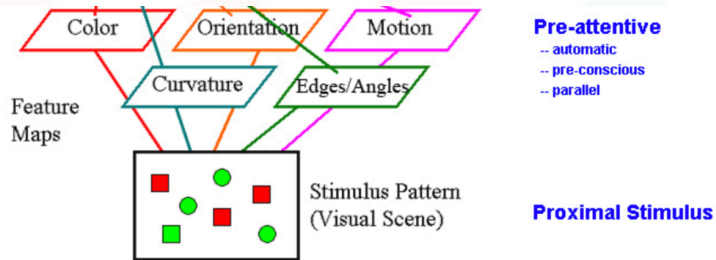


Parallel feature processing



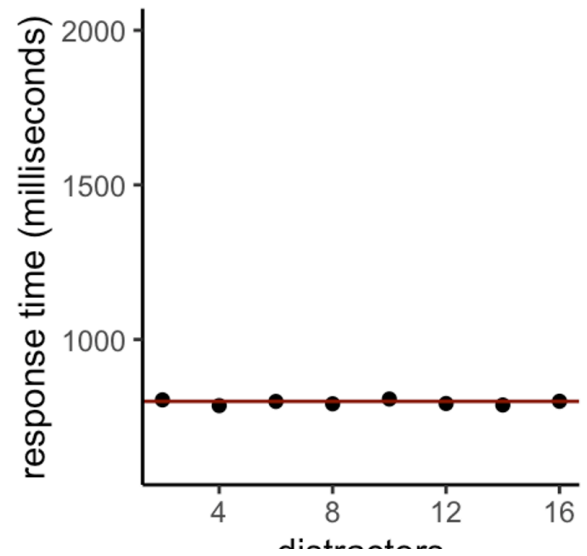
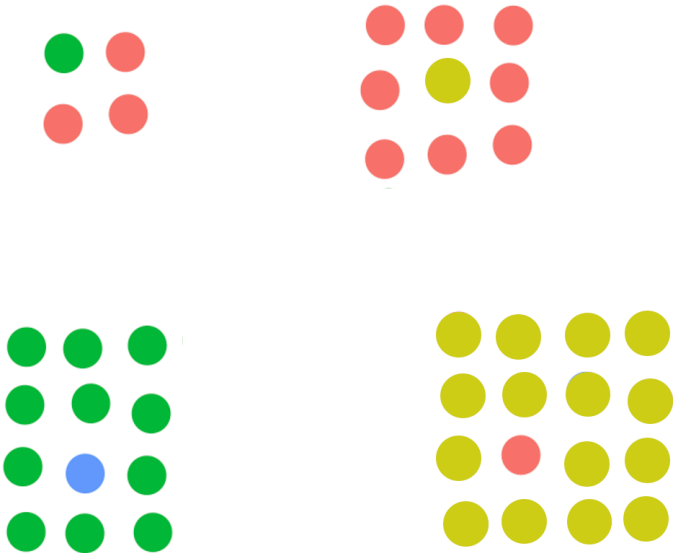
Features

- Feature *dimensions* include color, motion, orientation
 - Feature *values*: red, blue for color
 - Feature *values*: leftward, downward for motion



Parallel processing

Odd feature values are salient

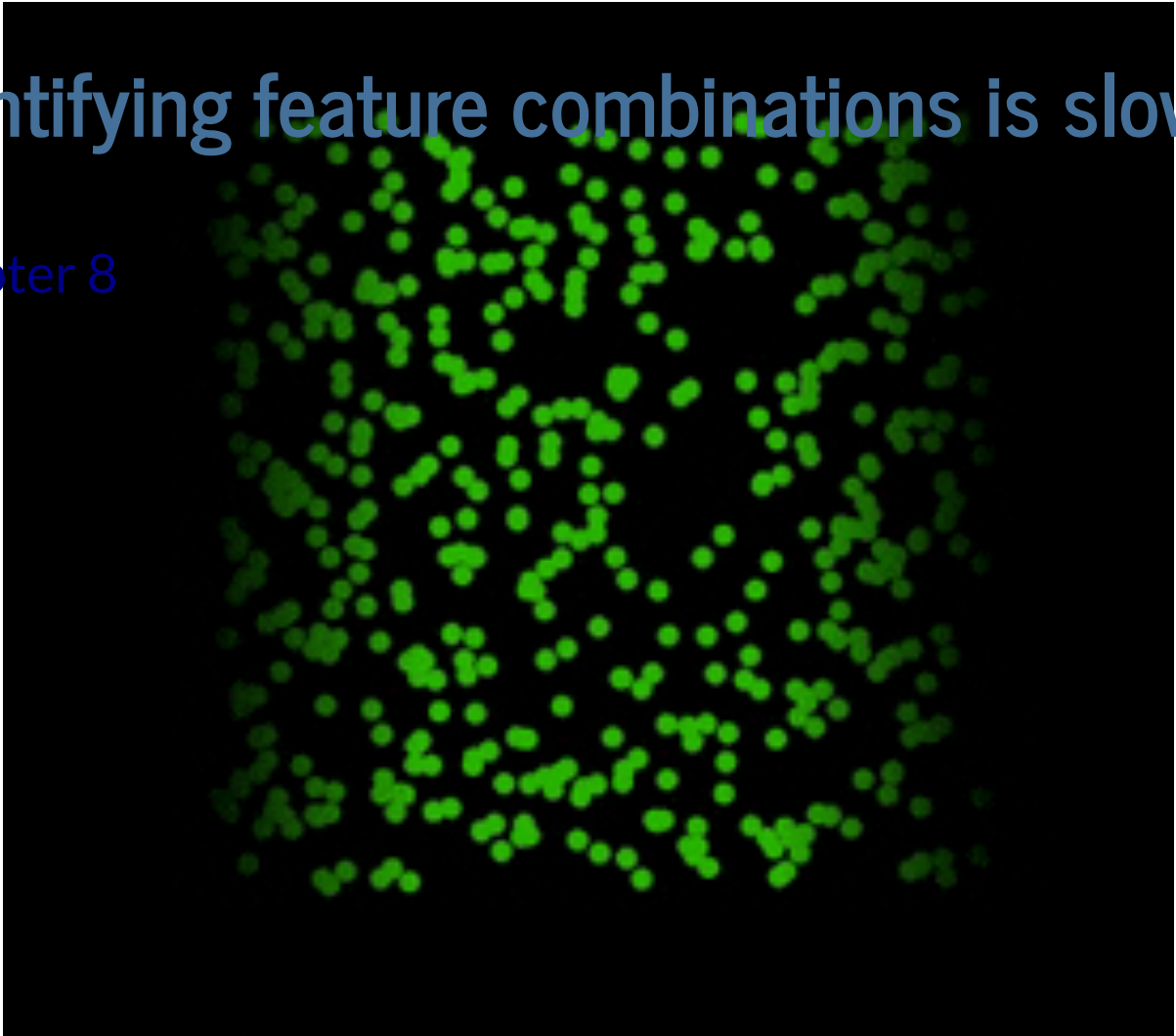


Feature processing

- *Within* a feature dimension, salience computation is massively parallel
- Identifying feature *combinations* is *slow*.

Identifying feature combinations is slow

Chapter 8



Identifying feature combinations is slow

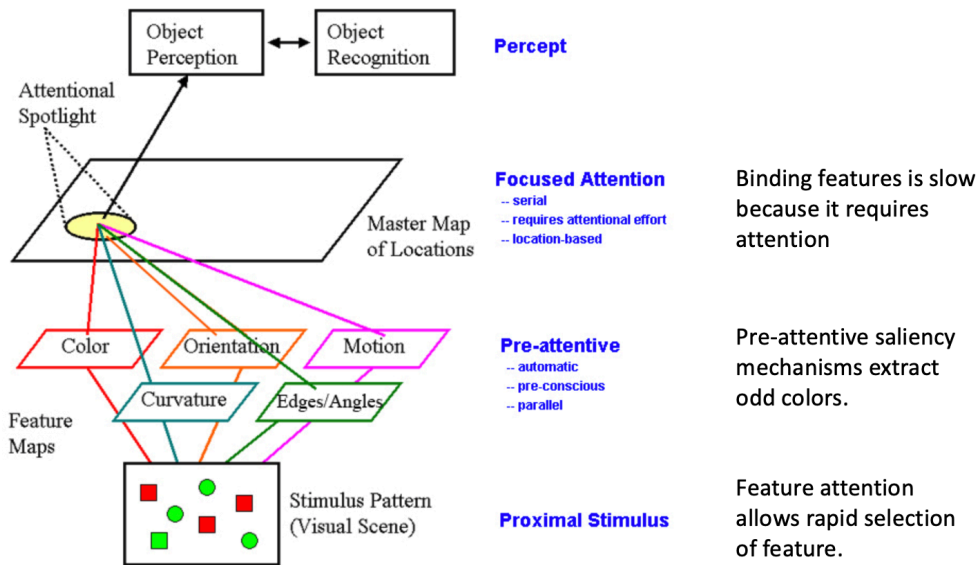
Possibly doesn't occur unless selective attention routes the features through the bottleneck.



3. Treisman's diagram and combining features

- Individual-feature salience is computed in parallel
- Flat search slopes (parallel) vs. steep (serial)
- Combining features from two different objects takes the longest, but even within an object takes awhile (both require focused attention)
- It feels like we see everything at once
- The mysterious non-selective pathway

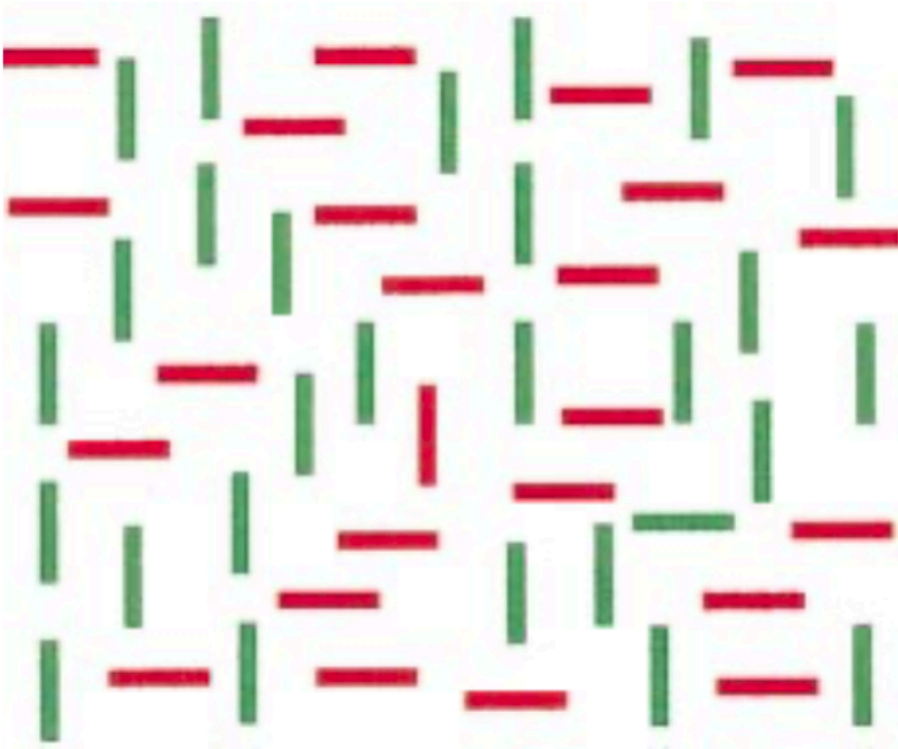
Feature integration theory of Treisman



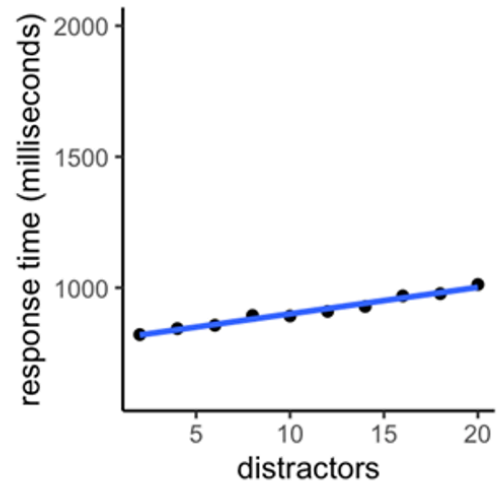
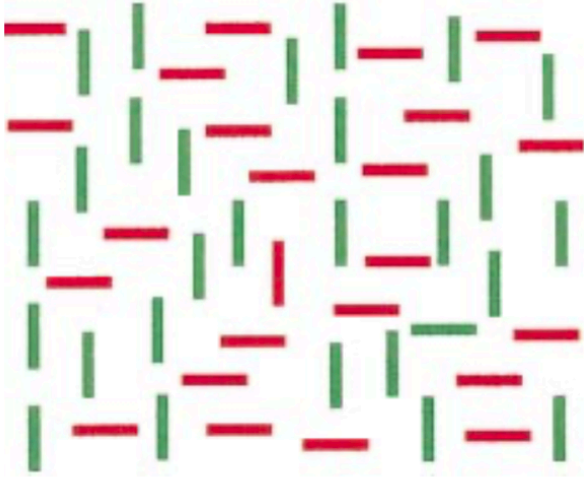
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Odd *pair* of features: *serial* processing

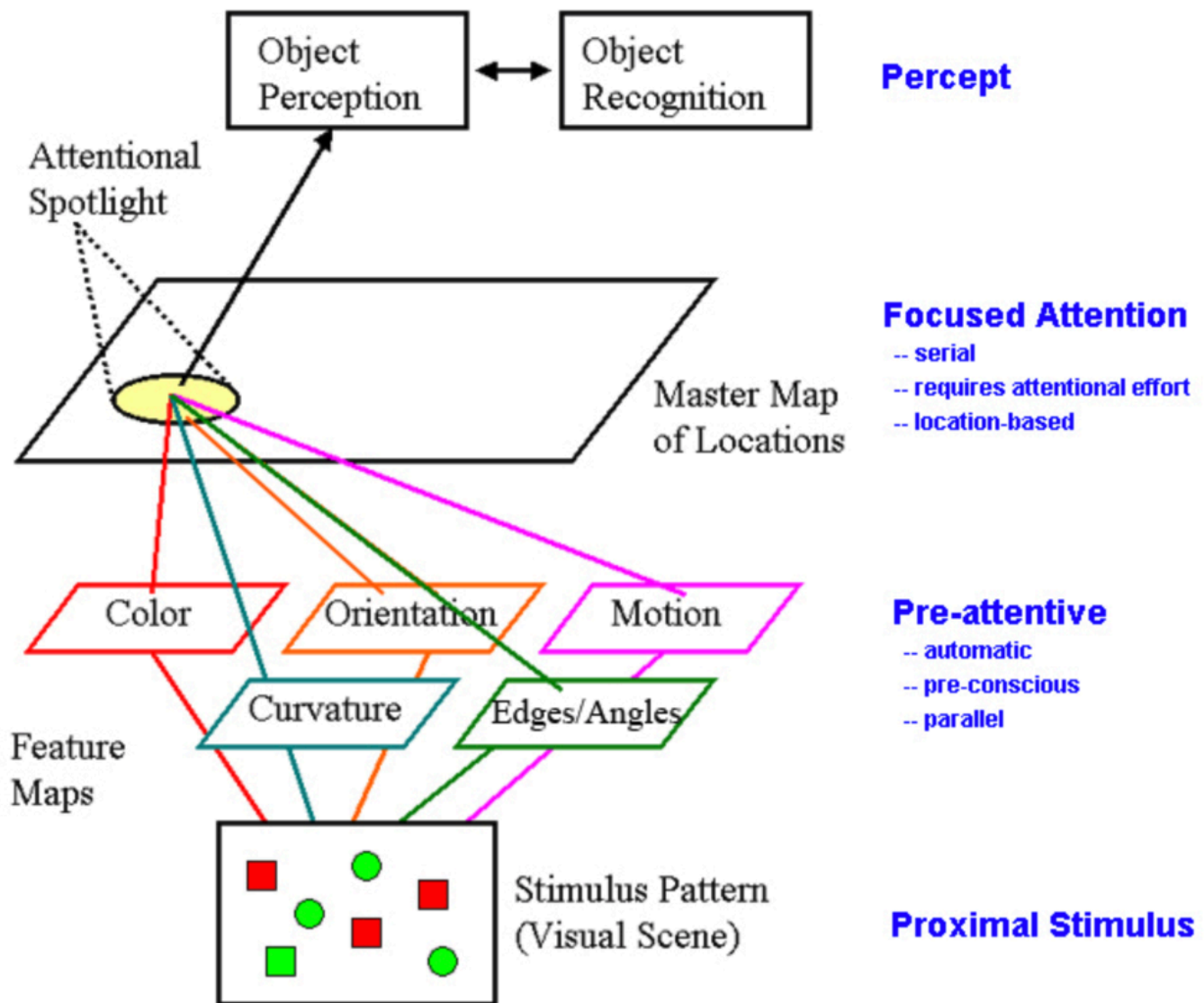
Get ready to look for the odd pair of features!



Odd feature pair: *serial* processing



Feature integration theory of Treisman



{.nonincremental .smallest background-image="images/brainBottomupTopdownTheeuwesFailing.png" background-opacity="0.08"}

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Thing(s) I didn't understand

Waiting for responses ...

